

CSE 360: Computer Interfacing
Department of Computer Science and Engineering
Brac University
Semester: Spring 2026
Assignment 2

1. You are tasked with building a **robotic arm control system** that interfaces with 2 IC 82C55, **16 limit switches** (to detect arm positions and safe limits), **8 motors** (each requiring two signals: Enable and Direction), **4 sensors** for obstacle detection each with 4 pins. Here, Port A of the first IC: reads 8 limit switches and Port B of the first IC: reads 8 more limit switches.
 - a. Design the complete connection diagram mapping for both 82C55 ICs.
 - b. Deduce the control words for Both ICs.
2. Explain how a character LCD internally processes and displays characters. Include details about RS, RW, EN pins.
3. Describe how an 8×8 LED matrix works.
4. Compare and contrast the working of **common cathode** and **common anode** 7-segment displays. Show how segment control logic differs in both cases.
5. Discuss the advantages and limitations of each display technology mentioned in question 2,3,4 - in terms of **Power consumption**, **interface complexity**, and **suitability** for battery-operated systems. *[You might use google and do some self study through it.]*